

CARCINOGENS & TERATOGENS

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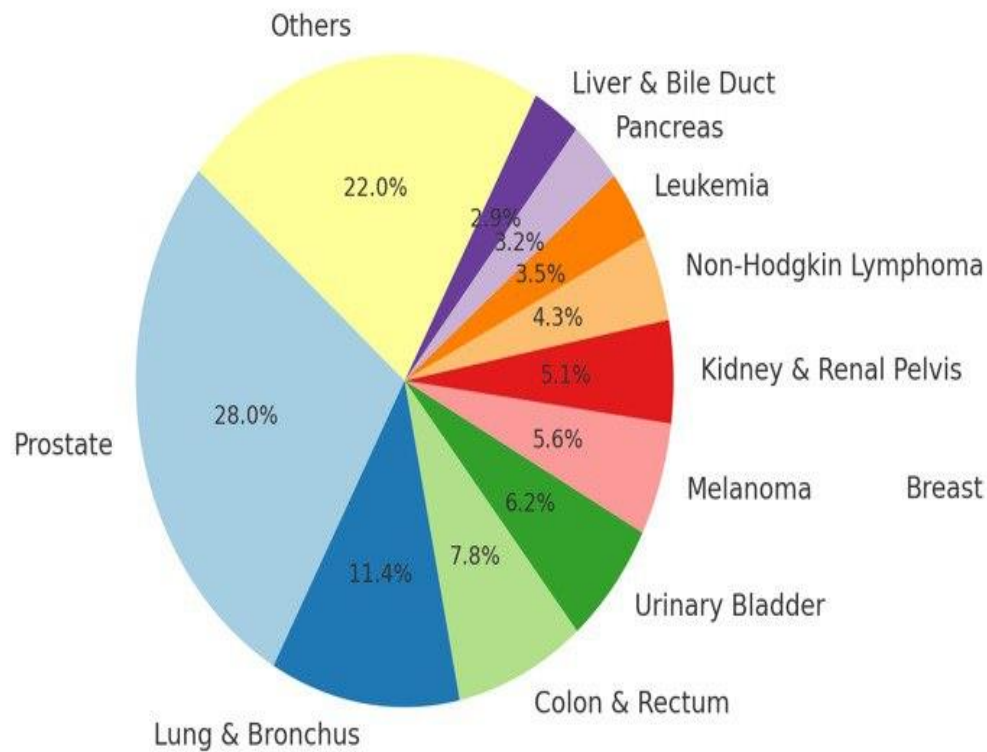
CARCINOGENS

Cancer

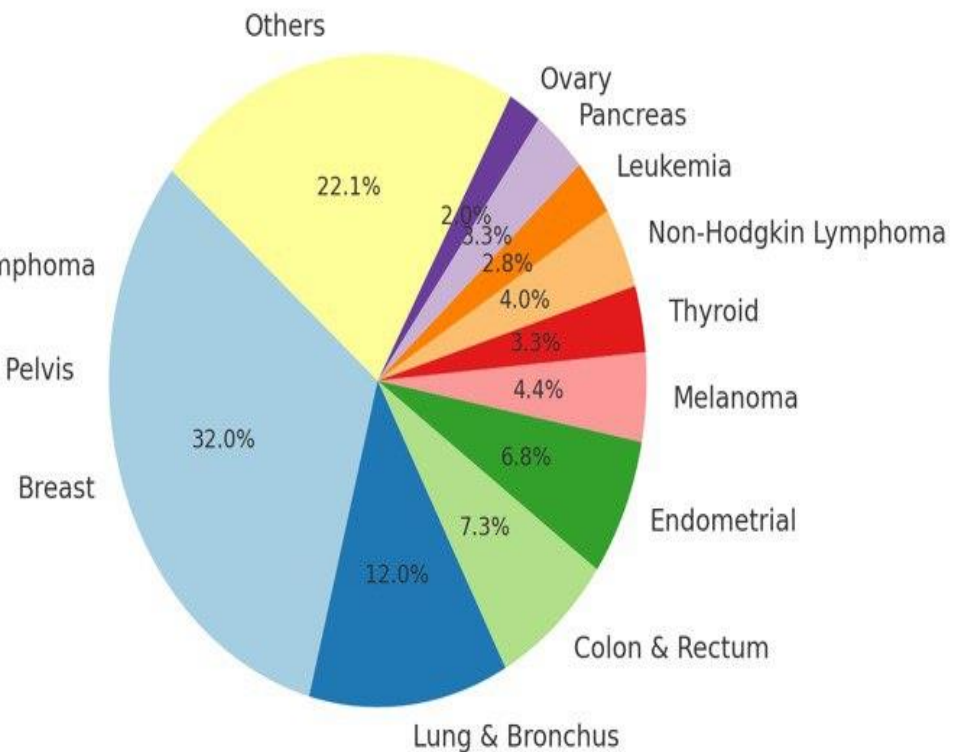
- Abnormal cells with uncontrolled cellular proliferation
- Disease characterized by *genomic mutation, modified gene expression, cell proliferation*, and **aberrant cell growth**
- It is the second most common cause of death in the Western countries. **MCC??**
- Cancer occurs mostly in *older people*, and the risk of developing cancer increases with age.
 - *~77 % of all cancers is diagnosed in people aged 55 and older*

Cancer statistics in the US for males and females

New Cancer Cases by Type (Males, 2024)



New Cancer Cases by Type (Females, 2024)



Terminologies

Term	Meaning
Neoplasm	An abnormal and uncontrolled growth of cells, which can be benign or malignant.
Tumor	A mass or lump of tissue formed by abnormal cell growth; can be benign or malignant.
Benign	A non-cancerous tumor that does not invade nearby tissues or spread to other parts of the body.
Malignant	A cancerous tumor that can invade nearby tissues and spread (metastasize) to other areas
Cancer	A disease caused by uncontrolled cell division, leading to malignant neoplasia/tumor that may spread.
Metastasis	The process by which cancer cells break away from the original tumor and spread to other parts of the body.
Carcinoma	A type of cancer that begins in epithelial cells , which line the body's organs and surfaces (e.g., lung, breast, colon).
Sarcoma	A type of cancer that originates in connective tissues such as bone, muscle, fat, or cartilage.
Carcinogens	Substances (e.g., tobacco smoke, radiation, chemicals) that can cause neoplasia.

Etiology of cancer

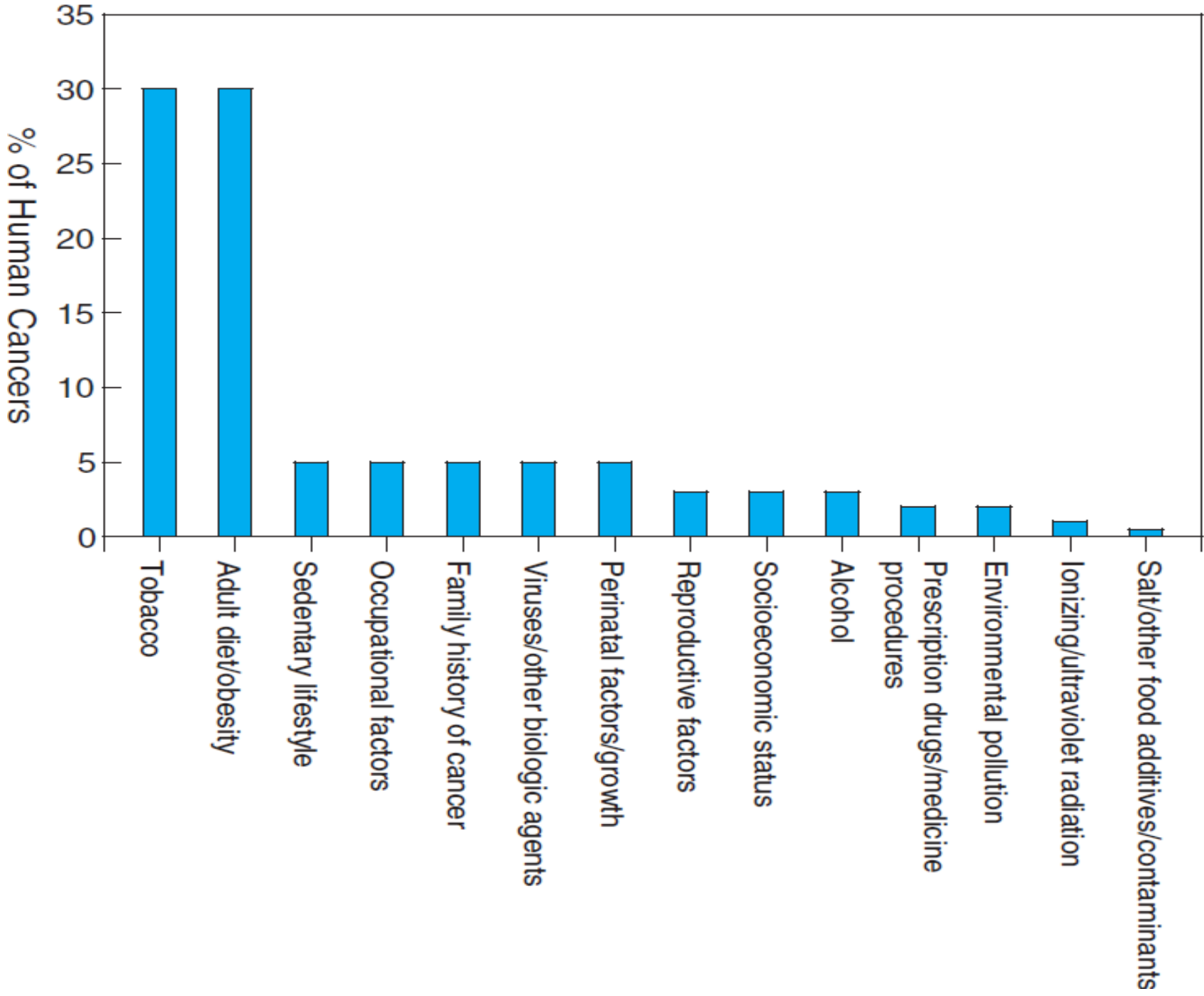
A normal cell turns into a cancer cell because of mutations in its DNA, which can be inherited or acquired

- ✓ Hereditary (E.g BRCA1 & 2)
- ✓ Tobacco
- ✓ Chemicals
- ✓ Drugs
- ✓ Radiations (UV, RT)
- ✓ Dietary factors
- ✓ Infections

Cancer susceptibility is determined by complex interactions between age, environment, and individual's genetic makeup

- 
- It is estimated from epidemiological studies that up to 90% of all human cancers are associated with the environment
 - <10% inherited genetic mutations
 - ❑ Environment applies to: chemicals, lifestyle including smoking, alcohol, diet, cultural and sexual behavior, occupation, natural and medical radiation.

Figure: Major causes of Human Cancers



➤ **Characteristics of Cancer cells** (*that distinguish them from normal cells*)

- *Uncontrolled proliferation*
- *Resistance to apoptosis*
- *Immortality*
- *Angiogenesis*
- *Immune system invasion*
- *Invasiveness*
- *Metastasis*

How do cancer cells differ normal cells

- Behave differently
- Are not stopped by neighbor cells (contact inhibition)
- Do not have programmed cell death
- Can spread (metastasize)

Carcinogenesis

- Normal cell becomes cancerous.
- Cancer is a disease of the genes
- Mutations (errors in genes) occur when cells divide
- Most mutations are repaired
- If the mutation is not repaired, mutated cells can survive and become malignant cancer

The cell cycle

- Can be divided in to two periods:
 - Interphase
 - ✓ The cell grows, accumulating nutrients needed for mitosis and duplicating its DNA
 - ✓ Resting phase but metabolism?
 - ✓ G1 phase, S phase, G2 phase
 - The mitotic (M) phase
 - ✓ division into two daughter cells
 - ✓ *Prophase, Metaphase, Anaphase, Telophase*

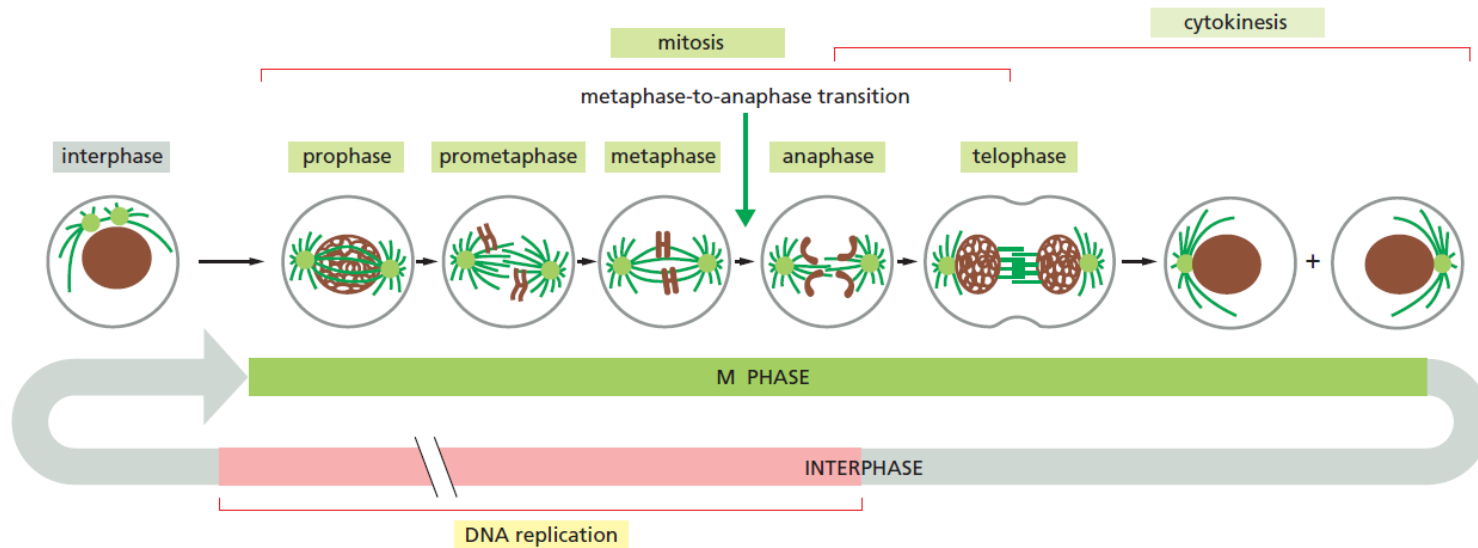


Figure 17–3 The events of eukaryotic cell division as seen under a microscope. The easily visible processes of nuclear division (mitosis) and cell division (cytokinesis), collectively called M phase, typically occupy only a small fraction of the cell cycle. The other, much longer, part of the cycle is known as interphase, which includes S phase and the gap phases (discussed in text). The five stages of mitosis are shown: an abrupt change in the biochemical state of the cell occurs at the transition from metaphase to anaphase. A cell can pause in metaphase before this transition point, but once it passes this point, the cell carries on to the end of mitosis and through cytokinesis into interphase.

- **G1**

- If extracellular conditions are unfavorable → cells delay progress through G1 and may even enter a specialized resting state known as **G0 (G zero)**,

- **G0** -in which they can remain for days, weeks, or even years before resuming proliferation.

- In the adult, most cells do not constantly divide; most spend a varying amount of time in a quiescent phase outside the cycle in the phase termed G_0

- G0 phase- RBCs, cardiac & skeletal muscle, neurons

- Some cells never enter G0 phase (e.g GIT epithelium, skin, bone marrow cells)

Cell cycle checkpoints

- ❖ Are the controlling mechanisms of cell growth

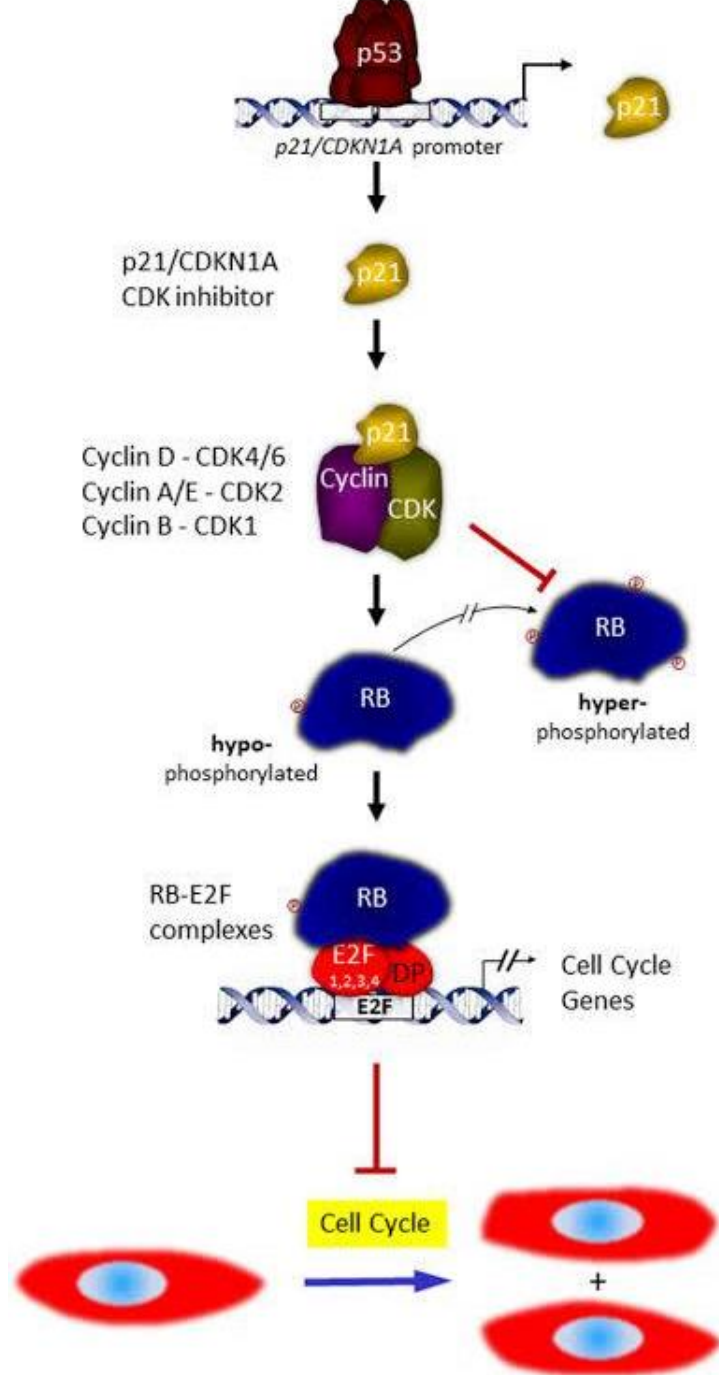
- ❖ **Two regulators**

- ❑ ***The stimulatory*** mitogenic growth factor

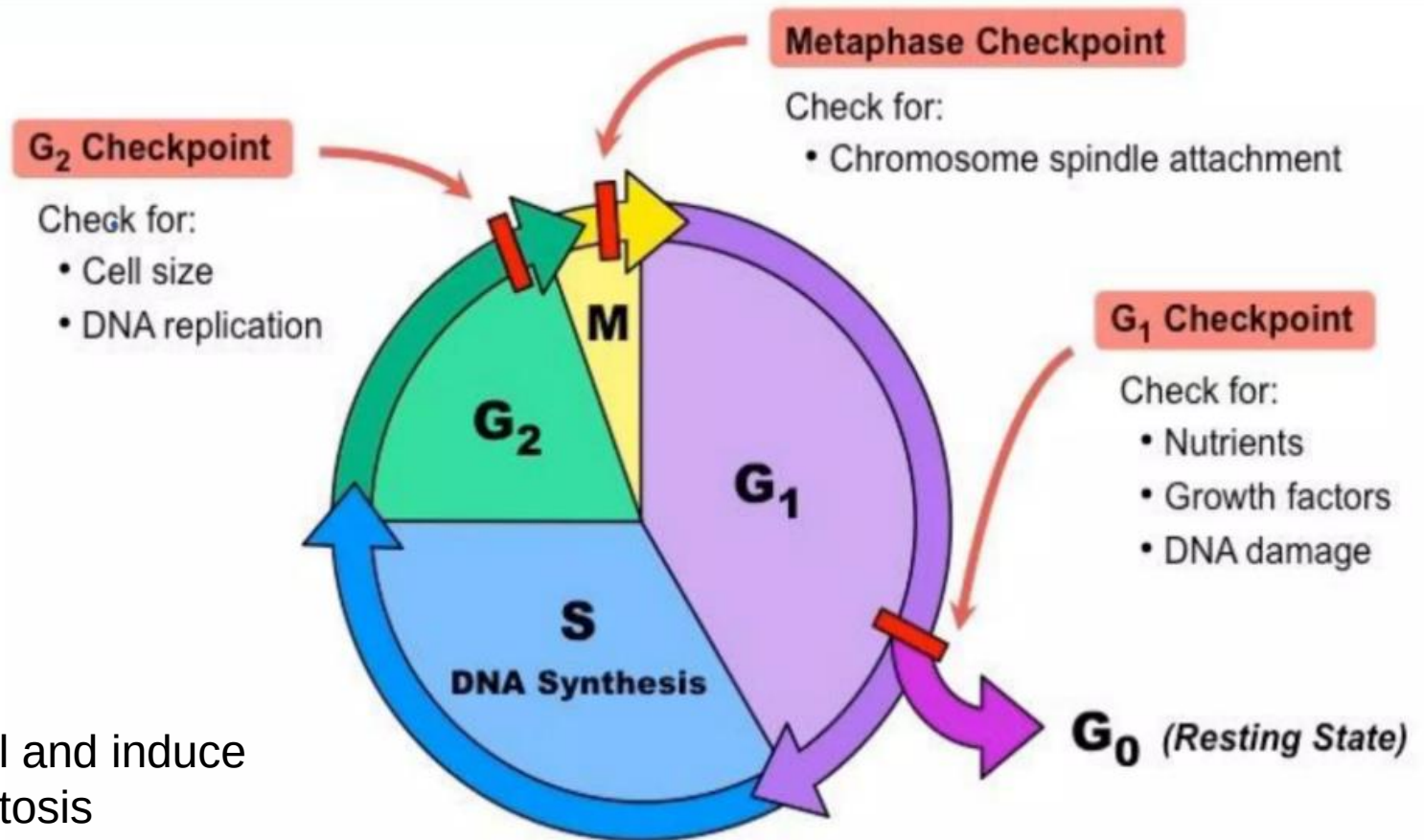
- **Cyclin** and **cyclin-dependent kinases (CDKs)**, which are expressed throughout the phases of the cell cycle.
- Cyclin-CDKs check allows cell cycle progression
 - **Cyclin** (structural protein) and
 - **Kinase** (enzyme).

- ❑ ***The inhibitory*** antimitogenic growth factors

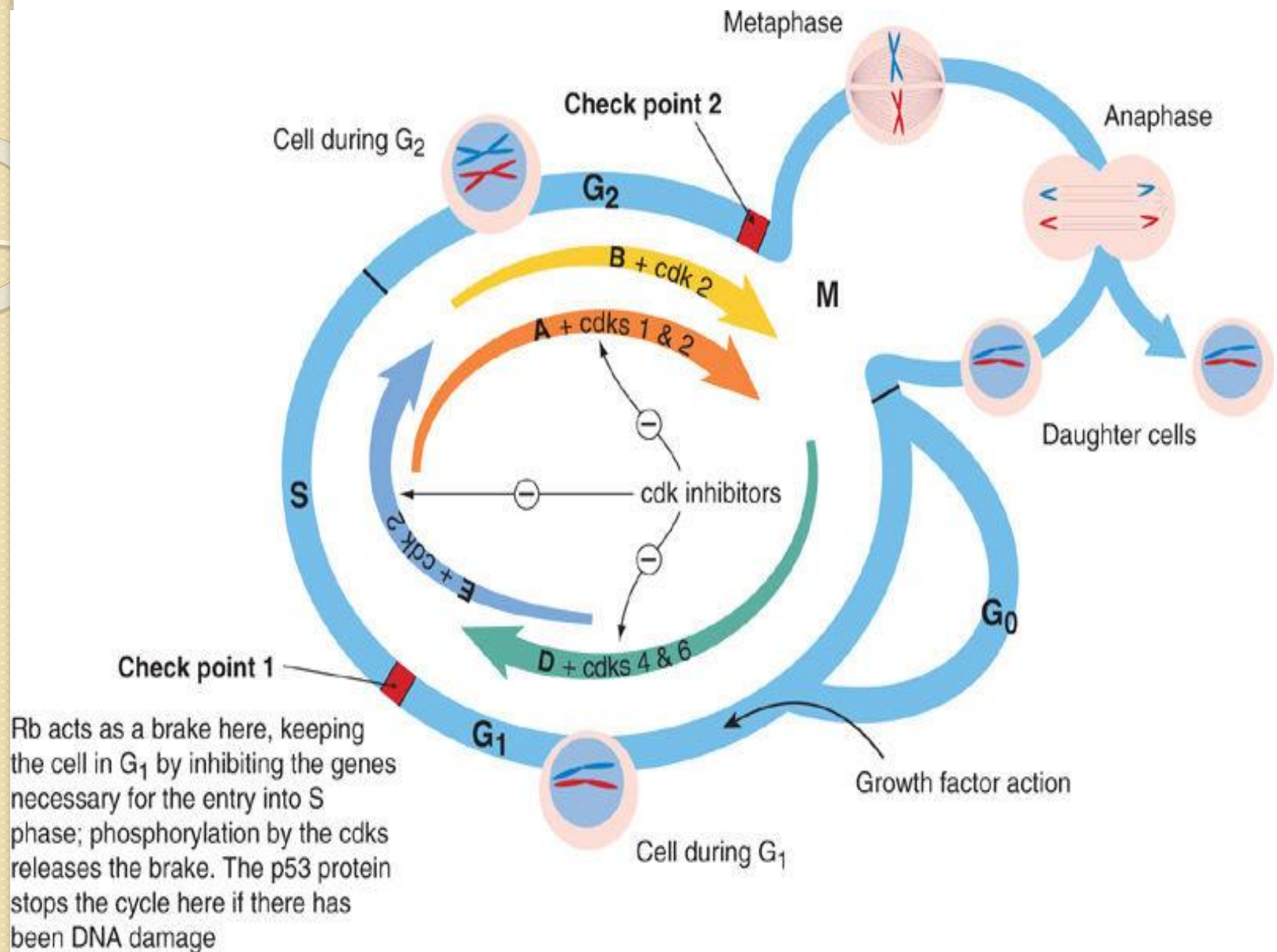
- p53/Rb and **cyclin dependent kinase inhibitors** cascade
- all arrest the cell cycle by inhibiting CDks



Checkpoints



P53-arrest cell and induce repair or apoptosis



Rang et al: Rang & Dale's Pharmacology, 7e
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- 
- The activity of the cdks is regulated by cdk inhibitors. If there is DNA damage, the products of the tumor suppressor gene *p53* stop the cycle at **check point 1**, allowing for **DNA repair**. If repair fails, apoptosis is initiated
 - *p53* is a protein activates the transcription of several genes, and synthesis of proteins that are CDKs inhibitors (p21, p16, and p27)
 - DNA damage can also result in the cycle being stopped at **check point 2** (check for new mutation).

- 
- The maintenance of normal cell numbers in tissues and organs requires that there should be a balance between the *positive regulatory forces* and the *negative regulatory forces*
 - **Inactivating** of p53/pRb, cyclin dependent kinase inhibitors cascade and/or enhanced cyclin or CDK activity, will lead to relentless *proliferation* in tumor cells – **carcinogenesis**



➤ Cancer is presumed to arise from the accumulation of multiple mutations in growth-regulatory genes

- The p53 tumor suppressor gene
- The proto-oncogene e.g. *ras*

The activation of proto-oncogenes to oncogenes

The inactivation of tumor suppressor genes

(e.g. p53 gene) ———**carcinogenesis**

Tumor-suppressor genes (TSG)- anti-oncogenes

- **p53** has been termed the “guardian of genome” because it controls a ***G1/S checkpoint***, regulates DNA repair, and apoptosis
- The *p53* gene is one of the most common tumor-suppressor genes, and mutations of *p53* may occur in up to *50% of all malignancies*
- The *loss of function of tumor suppressor genes* can be the critical event in carcinogenesis



➤ *Cellular changes occur in the genetic material of the cancer cell so that programmed cell death, or **apoptosis**, does not occur*

✓ **Loss of *p53* and over-expression of *bcl-2***

(oncogene - a generalized cell death suppressor gene) are two examples of changes within the cell that occur to result in enhanced cell survival by preventing apoptosis

P53 tumor suppressor protein



P53 activation



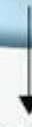
P53



BCL-2



BAX activation



Cytochrome c release



Caspase mediated apoptosis

Proto-oncogenes(PO)

- *are normal genes that **control cell division & survival***
- They encode proteins such as growth factors, receptors, signal transducers, and transcription factors (*e.g* HER2 gene, EGF, Ras gene)
- when proto-oncogenes are ***mutated*** or abnormally activated, they can become oncogenes

oncogenes

- Over a 100 oncogenes have been identified with approximately 30 oncogenes having a major role in human cancer...they are always on
- ***Mutations of proto-oncogenes***/oncogenes, such as the **ras genes** may result in their activation, leading to neoplastic transformation

List of some of oncogenes categorized by their function

i. Growth Factor Receptors

- EGFR (ERBB1) – Overexpressed in lung and brain cancers.
- HER2 (ERBB2) – Amplified in breast and gastric cancers.
- MET – Activated in renal and lung cancers.
- FGFR1 – Involved in lung and bladder cancers.
- PDGFRA – Implicated in gastrointestinal stromal tumors (GISTs).

ii. Signal Transduction Molecules

- KRAS – Mutated in pancreatic, colorectal, and lung cancers.
- HRAS – Associated with bladder cancer.
- NRAS – Found in melanoma and hematologic malignancies.
- BRAF – Common in melanoma and thyroid cancer.

iii. Chromosomal Translocations & Fusion Oncogenes

These oncogenes result from chromosome rearrangements, forming fusion proteins that drive cancer.

- BCR-ABL1 – Found in chronic myeloid leukemia (CML).

iv. Transcription Factors

These regulate gene expression, influencing cell proliferation and survival.

- MYC (c-MYC) – Overexpressed in Burkitt's lymphoma and breast cancer.
- N-MYC – Amplified in neuroblastoma.
- L-MYC – Associated with small-cell lung cancer.
- FOS – Plays a role in osteosarcoma and lymphomas.
- JUN – Involved in sarcomas.

v. Cell Cycle Regulators

These oncogenes promote cell cycle progression, leading to uncontrolled growth.

- CCND1 (Cyclin D1) – Overactive in breast cancer, lymphoma, and melanoma.
- CDK4 – Amplified in melanoma and glioblastoma.

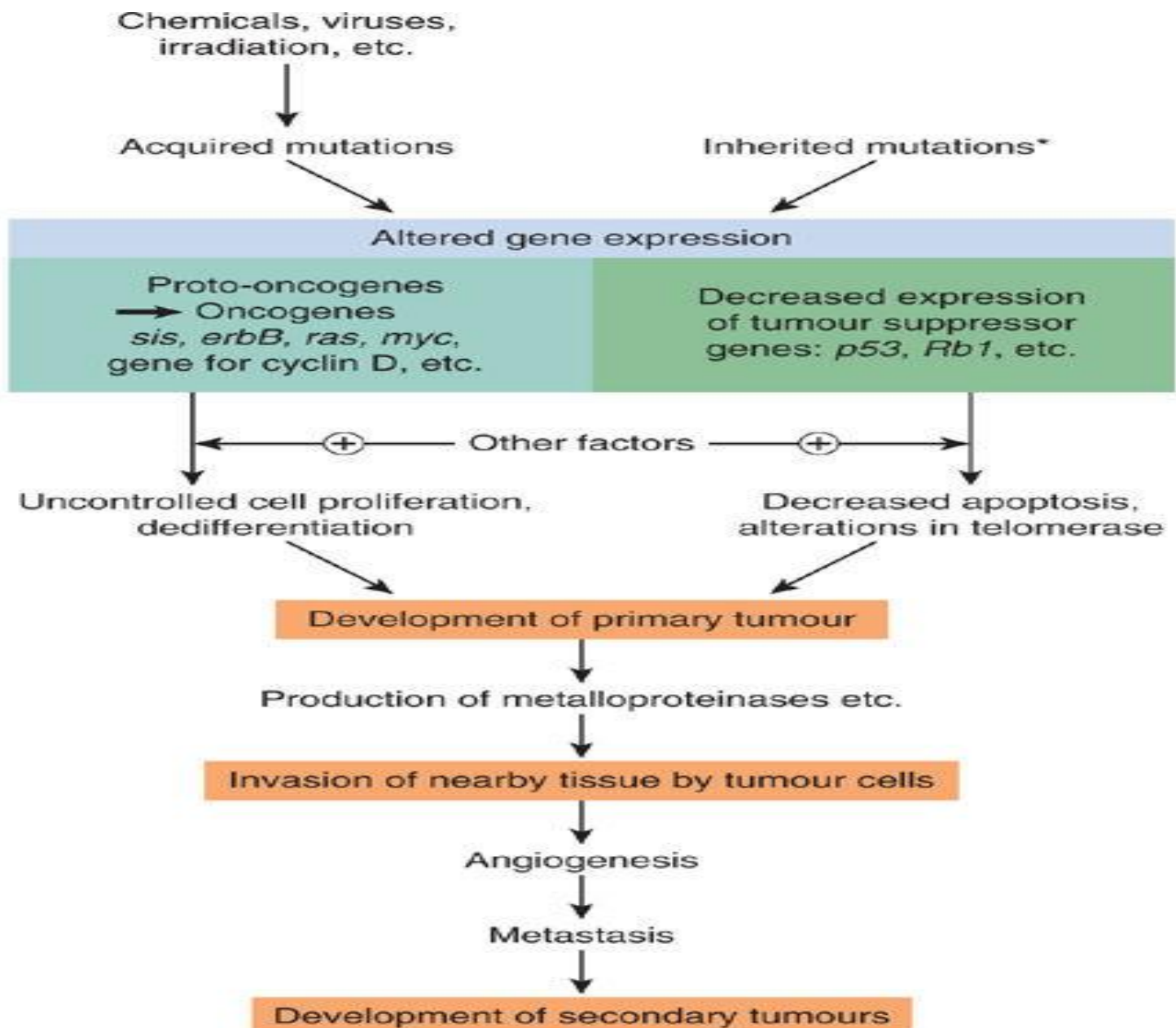
vi. Apoptosis Regulators

These oncogenes help cancer cells avoid programmed cell death (apoptosis).

- BCL-2 – Overexpressed in chronic lymphocytic leukemia (CLL).
- MDM2 – Inhibits p53, seen in sarcomas.

TABLE 85-1. Examples of Oncogenes and Tumor-Suppressor Genes

Gene	Function	Associated Human Cancer
Oncogenes		
<i>Genes for growth factors or their receptors</i>		
<i>EGFR</i> or <i>ERB-B1</i>	Codes for epidermal growth factor (EGFR) receptor	Glioblastoma, breast cancer, squamous carcinoma
<i>HER-2/neu</i> or <i>ERB-B2</i>	Codes for a growth factor receptor	Breast, salivary gland, prostate, bladder and ovarian cancer
<i>RET</i>	Codes for a growth factor receptor	Thyroid cancer
<i>Genes for cytoplasmic relays in stimulatory signaling pathways</i>		
<i>K-RAS</i>		Lung, ovarian, colon, pancreatic binding cancers
<i>N-RAS</i>	Code for guanine nucleotide-proteins with GTPase activity	Neuroblastoma, acute leukemia
<i>Genes for transcription factors that activate growth-promoting genes</i>		
<i>c-MYC</i>		Leukemia and breast, colon, gastric, and lung cancers
<i>N-MYC</i>		Neuroblastoma, small cell lung cancer, and glioblastoma
<i>Genes for cytoplasmic kinases</i>		
<i>BCR-ABL</i>	Codes for a nonreceptor tyrosine kinase	Chronic myelogenous leukemia
<i>Genes for other molecules</i>		
<i>BCL-2</i>	Codes for a protein that blocks apoptosis	Indolent B-cell lymphomas
<i>BCL-1</i> or <i>PRAD1</i>	Codes for cyclin D1, a cell cycle clock stimulator	Breast, head and neck cancers
<i>MDM2</i>	Protein antagonist of p53 tumor suppressor protein	Sarcomas
Tumor suppressor genes		
<i>Genes for proteins in the cytoplasm</i>		
<i>APC</i>	Step in a signaling pathway	Colon and gastric cancer
<i>NF-1</i>	Codes for a protein that inhibits the stimulatory Ras protein	Neurofibroma, leukemia, and pheochromocytoma
<i>NF-2</i>	Codes for a protein that inhibits the stimulatory Ras protein	Meningioma, ependymoma, and schwannoma
<i>Genes for proteins in the nucleus</i>		
<i>MTS1</i>	Codes for p16 protein, a cyclin-dependent kinase inhibitor	Involved in a wide range of cancers
<i>RB1</i>	Codes for the pRB protein, a master brake of the cell cycle	Retinoblastoma, osteosarcoma, and bladder, small cell lung, prostate and breast cancers
<i>p53</i>	Codes for the p53 protein, which can halt cell division and induce apoptosis	Involved in a wide range of cancers
<i>Genes for protein whose cellular location is unclear</i>		
<i>BRCA1</i>	DNA repair, transcriptional regulation	Breast and ovarian cancers
<i>BRCA2</i>	DNA repair	Breast cancer
<i>VHL</i>	Regulator of protein stability	Renal cell cancer
<i>MSH2, MLH1, PMS1, PMS2, MSH6</i>	DNA mismatch repair enzymes	Hereditary nonpolyposis colorectal cancer



DNA repair genes

- Code for proteins whose normal function is to ***correct errors*** that arise when cells duplicate their DNA prior to cell division.
- ***Mutations in DNA repair genes*** can lead to a failure in repair, which in turn allows subsequent mutations to accumulate

Stages of carcinogenesis

- Carcinogenesis is a complex multistage process,
- usually involving more than one **genetic change** as well as other, **epigenetic factors** (hormonal, co-carcinogen and tumor promoter effects, etc.) that do not themselves produce cancer but which increase the likelihood that the genetic mutation(s) will eventually result in cancer

Initiation

- Phenomenon of gene alteration, which may result from the interaction of *ultimate* carcinogens with DNA in the target cell
- Initiating event involves cellular genome – MUTATIONS

- ✓ Results from DNA mutations(e.g oncogenes/PO, TSG)
- ✓ Irreversible
- ✓ Initiated cells may remain ‘dormant’ *static nondividing* state
- ✓ *Could be deleted* through apoptotic mechanisms
- ✓ initiated *cell may proliferate*
- ✓ By itself **not sufficient** for neoplastic formation

Promotion

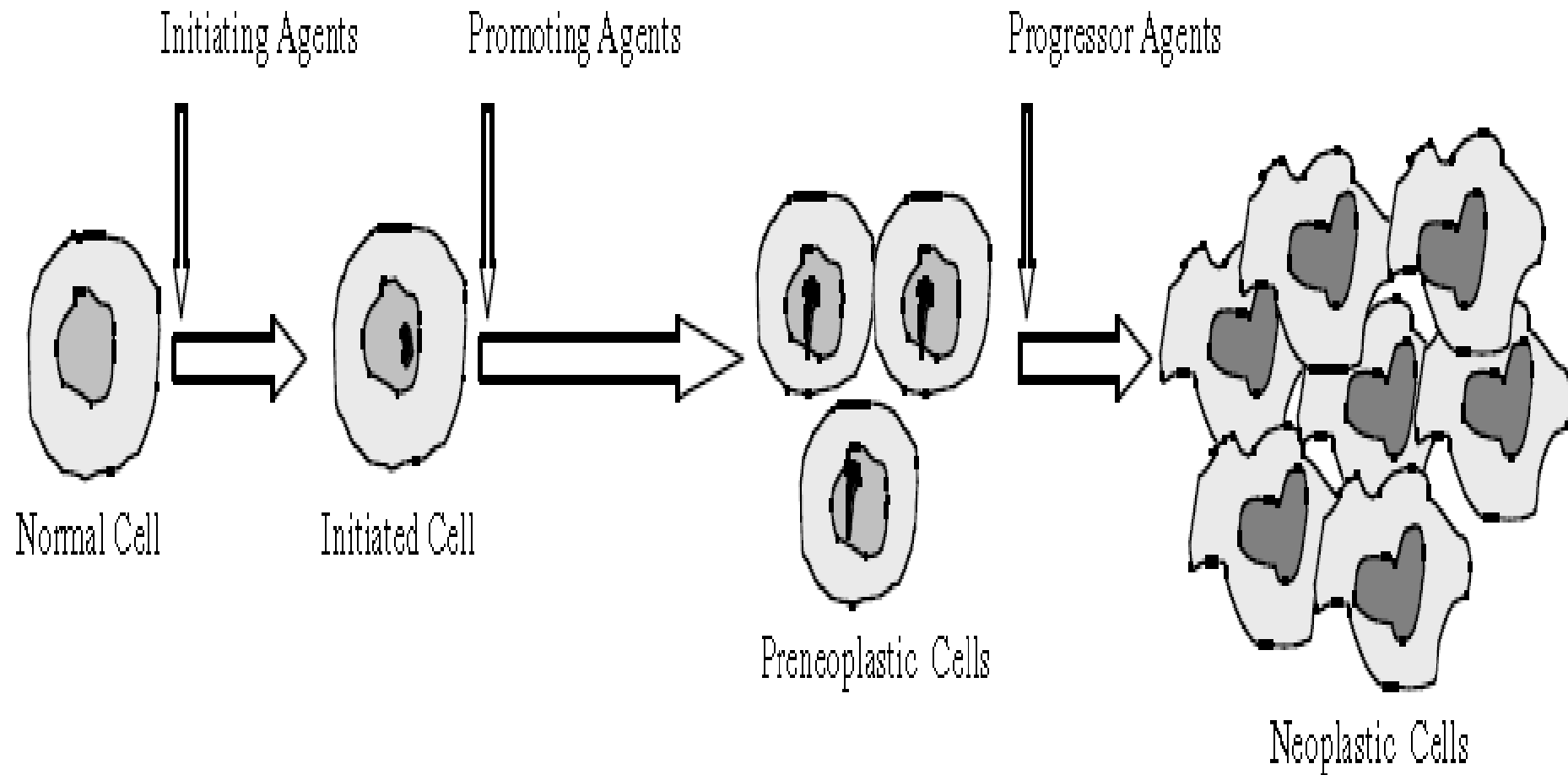
- A phenomenon of gene activation in which ***initiated cell*** becomes expressed through selection and ***clonal expansion***

- ✓ Results from ***clonal*** expansion of initiated cells
- ✓ Cell signaling and gene expression are altered, leading to increased ***cell mitogenesis*** and/or reduced ***apoptosis***
- ✓ Leads to the formation of **preneoplastic cells**
- ✓ Long duration and **reversible**, especially at earlier stage
- ✓ Agents capable of inducing tumor promotion - tumor promoters
- ✓ In contrast to **initiating agents**, tumor promoters do not directly interact with **DNA**. The intermediate stage of promotion does not appear to involve direct structural changes in the cell but rather depends on an *altered expression of genes*

Progression

- The last *irreversible*_stage of multistage carcinogenesis, which usually develops from the cells in the stage of promotion
- Characterized by *irreversibility*, genetic *instability*, faster growth, *invasion*, *metastasization*, and changes in the biochemical, metabolical and morphological characteristics of cells – for **malignant** phenotypes
- Leads to the formation of **benign or malignant** neoplasms
- Results from additional *genomic structural alterations*
- Agents that only cause the transition of a cell from the stage of promotion to that of progression are termed *progressor agents*

Multistage model carcinogenesis



Carcinogen

- *A carcinogen is an agent that causes or induces neoplasia OR*
- “A carcinogen is an **agent** whose administration to previously untreated animals leads to a **statistically significant increased incidence of neoplasms** of one or more histogenetic types as compared with the incidence in appropriate untreated animals”

- 
- Carcinogens can be ***chemicals, physical agents*** (such as *ultraviolet* light and gamma radiation), and ***biological agents*** (such as viruses)
 - Chemical carcinogens are the most commonly investigated carcinogenic agents
 - Chemical carcinogens that require metabolic activation to exert their carcinogenic effects are termed ***procarcinogens***, whereas their highly reactive metabolites are designated ***ultimate carcinogens***

Carcinogens: *can be divided into two major categories based on their response to in-vitro mutagenicity assay*

- Chemicals that induce cancer have been classified into
 - ✓ ***DNA-damaging agents (genotoxic)*** - mutagenic in in-vitro assays
 - Considered to produce permanent alterations in the genetic material of the host in vivo
 - ✓ ***Epigenetic agents (non-genotoxic)*** - not mutagenic in in-vitro assays
 - *Don't alter the primary sequence of DNA but are considered to **alter the expression** or repression of **certain genes** and/or to produce disturbance in signal transduction pathways that influence cellular events related to *proliferation, differentiation, or apoptosis**

DNA-damaging agents (genotoxic)

- ***i. Direct acting carcinogens*** are intrinsically reactive compounds - ***do not require metabolic*** activation

Examples include

- the alkyl alkanesulfonates such as methyl methane-sulfonate
- Propane sulfone, nitrogen and sulfur mustards
- Beta propiolactone
- Coal tar carcinogens including benzo(*a*)pyrene, pesticides (2-acetylaminofluorene)

- **ii. Indirect-acting carcinogens** require *metabolic activation* by cellular enzymes to form the ultimate carcinogenic species
 - ✓ Procarcinogen(Px)→Proximate carcinogen →Ultimate (Ut) carcinogen
 - ✓ E.g Aflatoxin B1 (Pr) → Aflatoxin B1 2,3 epoxide(Ut)
 - ✓ Examples
 - N-Nitrosoamines
 - Azo dyes, Mycotoxin
 - Aflatoxin B1, Carbamates etc...
- **iii. Radiation** and oxidative DNA damage can occur directly or indirectly
- **iv. Inorganic carcinogens** such as arsenic, chromium and nickel (oxidative stress, altered cell signaling, modulation of apoptosis, DNA methylation modification etc)

Epigenetic Agents

- Many epigenetic agents favor the proliferation of cells with an *altered genotype* (cells containing a mutated oncogene(s) and/or tumor suppressor gene(s)) and allow the *clonal expansion* of these *altered or “initiated” cells*

Table 8-11

Proposed Modes of Action for Selected Non-genotoxic Chemical Carcinogens

MODE OF ACTION	EXAMPLE
<i>Cytotoxicity</i>	Chloroform Melamine
<i>α2u-Globulin binding</i>	D-Limonene 1,4-Dichlorobenzene
<i>Receptor-mediated</i>	
CAR	Phenobarbital Toxaphene
PPAR α	Trichloroethylene Perchloroethylene Diethylhexylphthalate Fibrates (e.g., clofibrate)
AhR	2,3,7,8-Tetrachlorodibenzo- p-dioxin (TCDD) Polychlorinated biphenyls (PCBs) Polybrominated biphenyls (PBBs)
<i>Hormonal</i>	Biogenic amines Steroid and peptide hormones Diethylstilbestrol (DES) Phytoestrogens (Bisphenol-A) Tamoxifen Phenobarbital
<i>Altered methylation</i>	Phenobarbital Choline deficiency Diethanolamine
<i>Immunosuppression</i>	Atrazine Bisphenol A Phthalates
<i>Oxidative stress inducers</i>	Ethanol TCDD Lindane Dieldrin Acrylonitrile

Some Non-mutagenic Chemical Carcinogens

Benzene
Butylated hydroxyanisole
Chlorobenzilate
Chloroform
Clofibrate
Dieldrin
Diethylhexyl Phthalate
p,p-Dichlorodiphenyldichloroethylene
1,4 Dioxane
Nasal turbinate
Furfural
Lindane
Methapyrilene
Polychlorinated biphenyls
Reserpine
Saccharin
2,3,7,8-Tetrachlorodibenzo-p-dioxin
Trichloroethylene

Classification of Chemical Carcinogens in Relation to Their Action on One or More Stages of Carcinogenesis

- **Initiating agent** (incomplete carcinogen): a chemical capable only of initiating cells (e.g/ benzo(a)pyrene, UVB, nitrosamines, viruses)
- **Promoting agent**: a chemical capable of causing the expansion of initiated cell clones (e.g. tetradecanoyl phorbol acetate (TPA), phenobarbital, 2,3,7,8-tetracholorodibenzo-p-dioxin (TCDD), and cholic acid, phenobarbital, androgens and estrogens, natural and synthetic, are effective promoting agents)
- **Progressor agent**: a chemical capable of converting an initiated cell or a cell in the stage of promotion to a potentially malignant cell (e.g. hydrogen peroxide, arsenic salts, hydroxyurea, and organic peroxides, such as benzoyl peroxide, benzene)
- **Complete carcinogen**: a chemical possessing the capability of inducing cancer from normal cells, usually possessing properties of initiating, promoting, and progressor agents (e.g. aflatoxin, tobacco smoking)

Table 8-29

**IARC Classification of the Evaluation of
Carcinogenicity for Human Beings**

GROUP	EVIDENCE
1. Agent is carcinogenic to humans	Human data strong Animal data strong
2A. Agent is probably carcinogenic to humans	Human epidemiology data suggestive Animal data positive
2B. Agent is possibly carcinogenic to humans	Human epidemiology data weak Animal data positive
3. Agent is not classifiable as to carcinogenicity to humans	Human and animal data inadequate
4. Agent is probably not carcinogenic to humans	Human and animal data negative

Table 12.3 List of Agents, Substances, Mixtures, and Exposure Circumstances Known to be Human Carcinogens

Aflatoxins

4-Aminobiphenyl

Arsenic and certain arsenic compounds

Asbestos

Azathioprine

Benzene

Benzidine

Beryllium and certain beryllium compounds

N,N-bis-(2-Chloroethyl)-2-naphthylamine (chlornaphazine)

Bis(chloromethyl) ether and chloromethyl methyl ether

1,4-Butanediol dimethylsulfonate (Myleran®)

Cadmium and certain cadmium compounds

Chlorambucil

1-(2-Chloroethyl)-3-(4-methylcyclohexyl)-1-nitrosourea (MeCCNU)

Chromium and certain chromium compounds

Cyclophosphamide

Table 12.3 (continued)

Cyclosporin A (cyclosporin)

Diethylstilbestrol

Epstein-Barr virus

Erionite

Estrogen therapy

Estrogens, nonsteroidal

Estrogens, steroidal

Ethylene oxide

Etoposide in combination with cisplatin and bleomycin

Helicobacter pylori

Hepatitis B virus (chronic infection)

Hepatitis C virus (chronic infection)

Herbal remedies containing plant species of the genus *Aristolochia*

Human immunodeficiency virus, type 1

Human papillomavirus, type 16

Human papillomavirus, type 18

Human T-cell lymphotropic virus, type 1

Melphalan

Methoxsalen with ultraviolet A therapy (PUVA)

MOPP and other combined chemotherapy including alkylating agents

Mustard gas

Table 12.3 (continued)

2-Naphthylamine

Neutrons

Nickel compounds

Opisthorchis viverrini

Oral contraceptives

Radionuclides α -particle emitting

Radionuclides β -particle emitting

Radon

Schistosoma haematobium

Silica

Solar radiation

Talc containing asbestiform fibers

Table 12.3 (*continued*)

Tamoxifen
2,3,7,8-Tetrachlorodibenzo-*para*-dioxin
Thiotepa [*tris*(1-aziridiny)phosphine sulfide]
Thorium dioxide
Treosulfan
Vinyl chloride
X and gamma(γ) Radiation

Mixtures

Alcoholic beverages
Analgesic mixtures containing phenacetin
Betel quid with tobacco
Coal tar and coal pitches
Mineral oils
Salted fish
Shale oils
Soots
Tobacco smoke and tobacco smokeless products
Wood dust

Table 12.3 (continued)

Exposure circumstances

Aluminium production
Auramine manufacture
Boot and shoe manufacture and repair
Coal gasification
Coke gasification
Furniture and cabinet making
Haematite mining with exposure to radon
Iron and steel founding
Isopropanol manufacture
Manufacture of magenta
Painter
Rubber industry
Strong inorganic acid mists containing sulfuric acid

Carcinogenic Factors Associated with Lifestyle

CHEMICAL(S), PHYSIOLOGIC CONDITION, OR NATURAL PROCESS	ASSOCIATED NEOPLASM(S)	EVIDENCE FOR CARCINOGENICITY
Alcoholic beverages	Esophagus, liver, oropharynx, and larynx	Sufficient
Aflatoxins	Liver	Sufficient
Betel chewing	Mouth	Sufficient
Dietary intake (fat, protein, calories)	Breast, colon, endo- metrium, gallbladder	Sufficient
Reproductive history		
1. Late age at 1st pregnancy	Breast	Sufficient
2. Zero or low parity	Ovary	Sufficient
Tobacco smoking	Mouth, pharynx, larynx, lung, esophagus, bladder	Sufficient

Exposures to Chemical Carcinogens in the Workplace

AGENT	INDUSTRIES AND TRADES WITH PROVED EXCESS CANCERS AND EXPOSURE	PRIMARY AFFECTED SITE
<i>Established</i>		
Para-aminodiphenyl	Chemical manufacturing	Urinary bladder
Asbestos	Construction, asbestos mining and milling, production of friction products and cement	Pleura, peritoneum, bronchus
Arsenic	Copper mining and smelting	Skin, bronchus, liver
Alkylating agents (mechloro-ethamine hydrochloride and bis[chloromethyl]ether)	Chemical manufacturing	Bronchus
Benzene	Chemical and rubber manufacturing, petroleum refining	Bone marrow
Benzidine, beta-naphthylamine, and derived dyes	Dye and textile production	Urinary bladder
Chromium and chromates	Tanning, pigment making	Nasal sinus, bronchus
Isopropyl alcohol manufacture	Chemical manufacturing	Cancer of paranasal sinuses
Nickel	Nickel refining	Nasal sinus, bronchus
Polynuclear aromatic hydrocarbons (from coke, coal tar, shale, mineral oils, and creosote)	Steel making, roofing, chimney cleaning	Skin, scrotum, bronchus
Vinyl chloride monomer	Chemical manufacturing	Liver
Wood dust	Cabinetmaking, carpentry	Nasal sinus

Carcinogenic Risks of Chemical Agents Associated with Medical Therapy and Diagnosis

CHEMICAL OR DRUG	ASSOCIATED NEOPLASMS	EVIDENCE FOR CARCINOGENICITY
Alkylating agents (cyclophosphamide, melphalan)	Bladder, leukemia	Sufficient
Inorganic arsenicals	Skin, liver	Sufficient
Azathioprine (immunosuppressive drugs)	Lymphoma, reticulum cell sarcoma, skin, Kaposi's sarcoma (?)	Sufficient
Chlornaphazine	Bladder	Sufficient
Chloramphenicol	Leukemia	Limited
Diethylstilbestrol	Vagina (clear cell carcinoma)	Sufficient
Estrogens		
Premenopausal	Liver cell adenoma	Sufficient
Postmenopausal	Endometrium	Limited
Methoxypsoralen with ultraviolet light	Skin	Sufficient
Oxymetholone	Liver	Limited
Phenacetin	Renal pelvis (carcinoma)	Sufficient
Phenytoin (diphenylhydantoin)	Lymphoma, neuro- blastoma	Limited
Thorotrast	Liver (angiosarcoma)	Sufficient

TERATOGEN



Children of thalidomide
(CNN/file)



- 
- **Teratogen**- an agent that interferes with normal embryonic or fetal development, causing, congenital anomalies/structural *malformations* or birth defects.
 - **Teratology** is a specialized area of embryology. It is the study of the etiology of abnormal development (*the study of birth defects*)

Etiology of teratogenesis

- Genetic – error in genetic programming
- Environmental –interference with embryogenesis and fetal development
 - Drugs, chemicals, radiation, infection(TORCH)

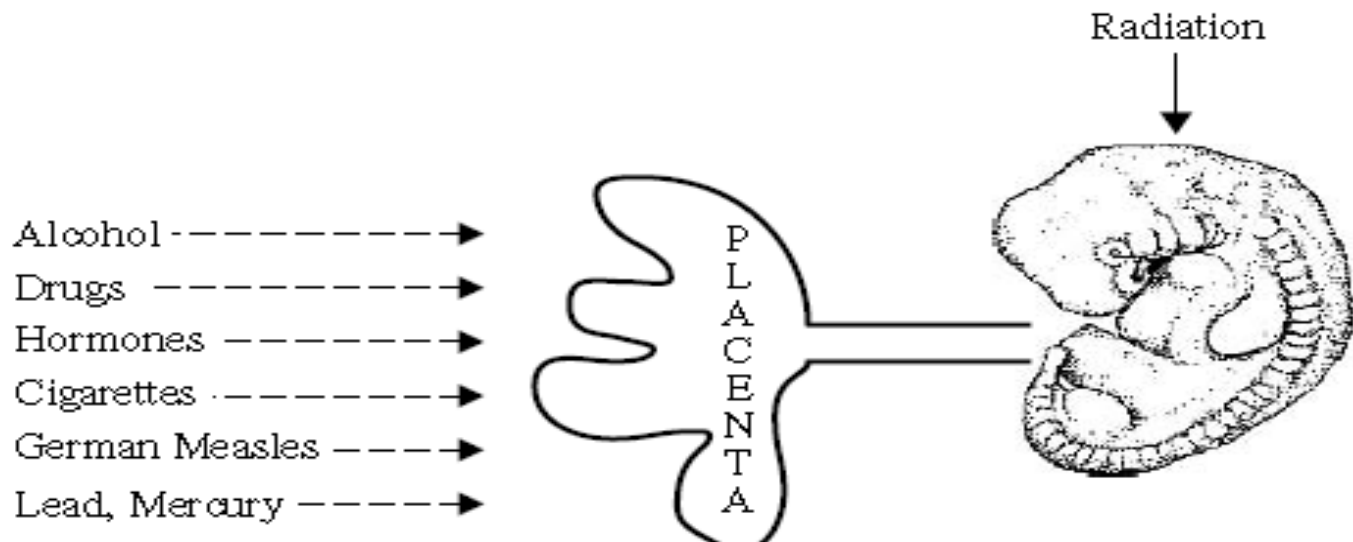


Table 10-3

Wilson's General Principles of Teratology

- I. Susceptibility to teratogenesis depends on the **genotype of the conceptus** and the manner in which this interacts with adverse environmental factors.
- II. Susceptibility to teratogenesis varies with the **developmental stage** at the time of exposure to an adverse influence.
- III. Teratogenic agents act in specific ways (**mechanisms**) on developing cells and tissues to initiate sequences of abnormal developmental events (pathogenesis).
- IV. The access of adverse influences to developing tissues depends on the **nature of the influence (agent)**.
- V. The four manifestations of deviant development are death, malformation, growth retardation, and functional deficit.
- VI. **Manifestations** of deviant development increase in frequency and degree as **dosage increases**, from the no effect to the totally lethal level.

SOURCES: From Wilson (1959, 1973).

❖ Critical Periods of Development and Susceptibility to Toxicity

➤ Susceptibility to teratogen varies by **gestation age**

✓ **pre-embryonic period** (from fertilization to early post – implantation)

- 1-2wks of GA
- Prior to organ differentiation
- Not usually associated with teratogenesis
- Here, effect of teratogen is all or none effect

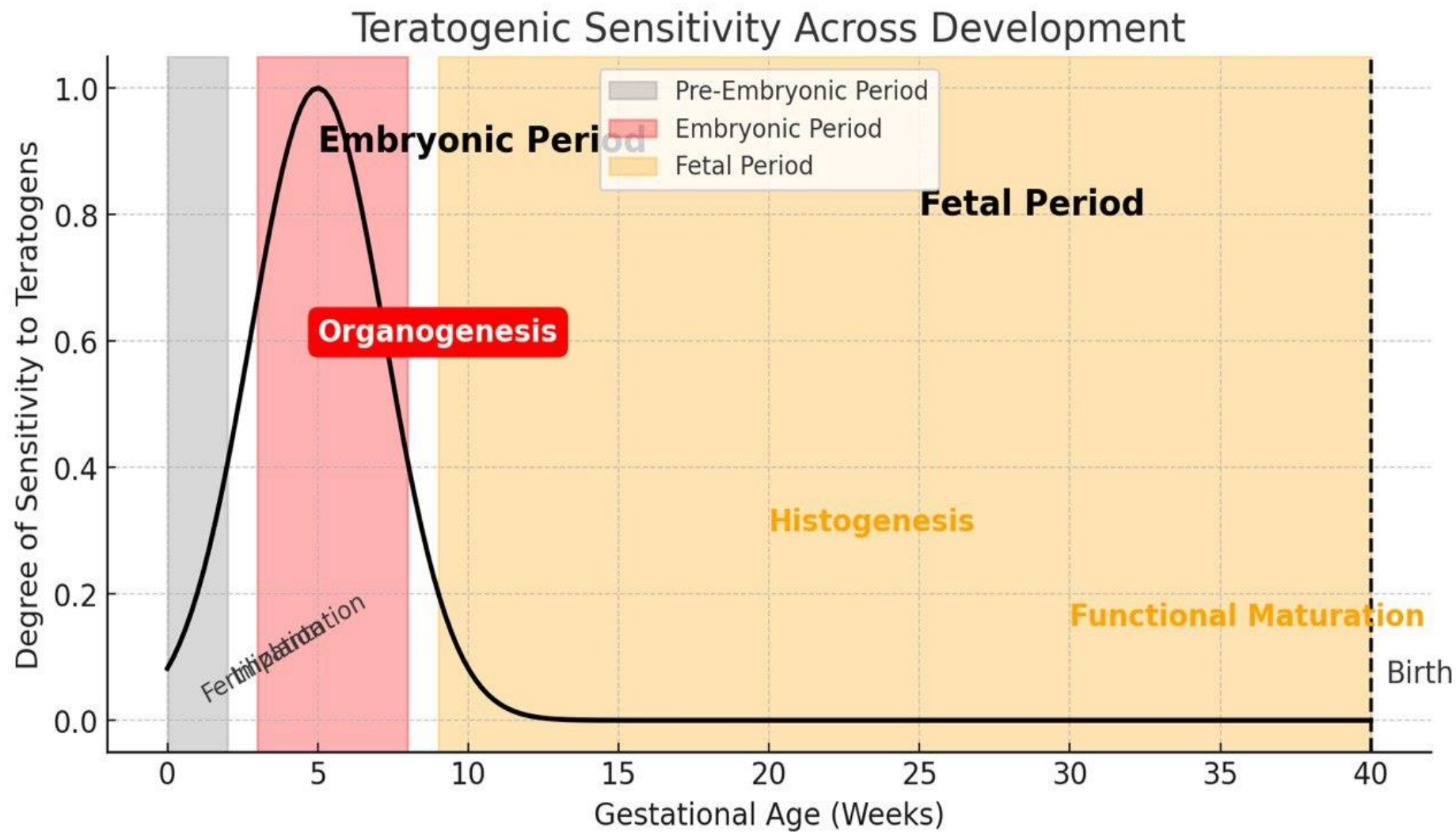


✓ **Embryonic period**

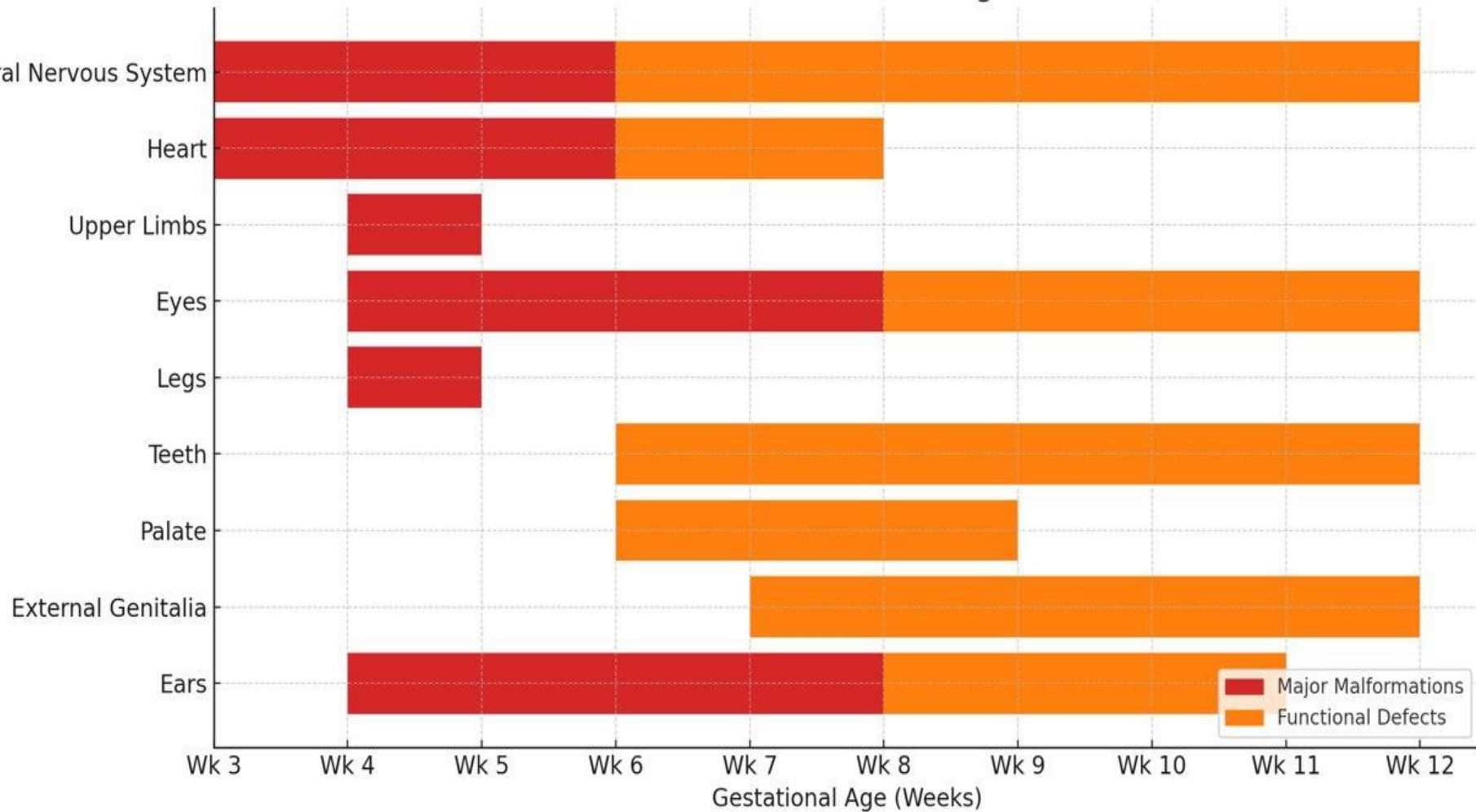
- Weeks 3-8
- *Gastrulation, organogenesis*
- Period of major organ development
- Greater risk of malformation; morphological defects

✓ **The fetal period:**

- > 9 weeks until birth
- Fetal growth and maturation of formed organs
- CNS and some organs continue to develop e.g. eyes, ears, teeth, external genitalia
- Abnormalities also occurs e.g. ethanol
- Functional disorders, growth retardation, *deformations*, carcinogenesis



Critical Periods for Teratogenesis



Dose – response relationship and threshold

- The major effects of prenatal exposure - are *embryo lethality, malformations, and growth retardation*
- The relationship between these effects is complex and varies with the *type of agent, the time of exposure, and the dose*
 - For some agents a continuum of increasing toxicity, with low dosages producing growth retardation and increasing dosages producing malformations and then lethality
 - Malformations and/or death can occur in the absence of any effect on intrauterine growth, but this is **unusual**

- Likewise, **growth retardation** and **embryo lethality** can occur without malformations. This is considered embryotoxic or embryolethal but not teratogenic.

Threshold dose

- All teratogens have **threshold doses**
- Threshold dose is always less than teratogenic dose
- The threshold will depend on the **specific stage** of development and the **organ system** affected

Teratogenic dose

- The dose causing a **malformation** may be below, equal to, or above the dose causing **acute toxicity**
- **The higher the dose the more severe the malformation but**
- **Timing** is crucial (small dose vs large dose)

Wilson's proposed mechanisms of teratogenesis

- ✓ Altered nucleic acid synthesis
- ✓ Altered mitosis,
- ✓ Gene mutation and chromosomal aberration
- ✓ Enzyme inhibition
- ✓ Lack of precursors and coenzymes,...antifolate
- ✓ Decreased energy supplies,
- ✓ Altered membrane characteristics

such cellular insults may quickly trigger ***pathogenetic responses*** in the embryo (E.g reduced cell proliferation, altered cell–cell interactions and signaling, reduced biosynthesis, inhibition of morphogenesis, or mechanical disruption of development, cell death)

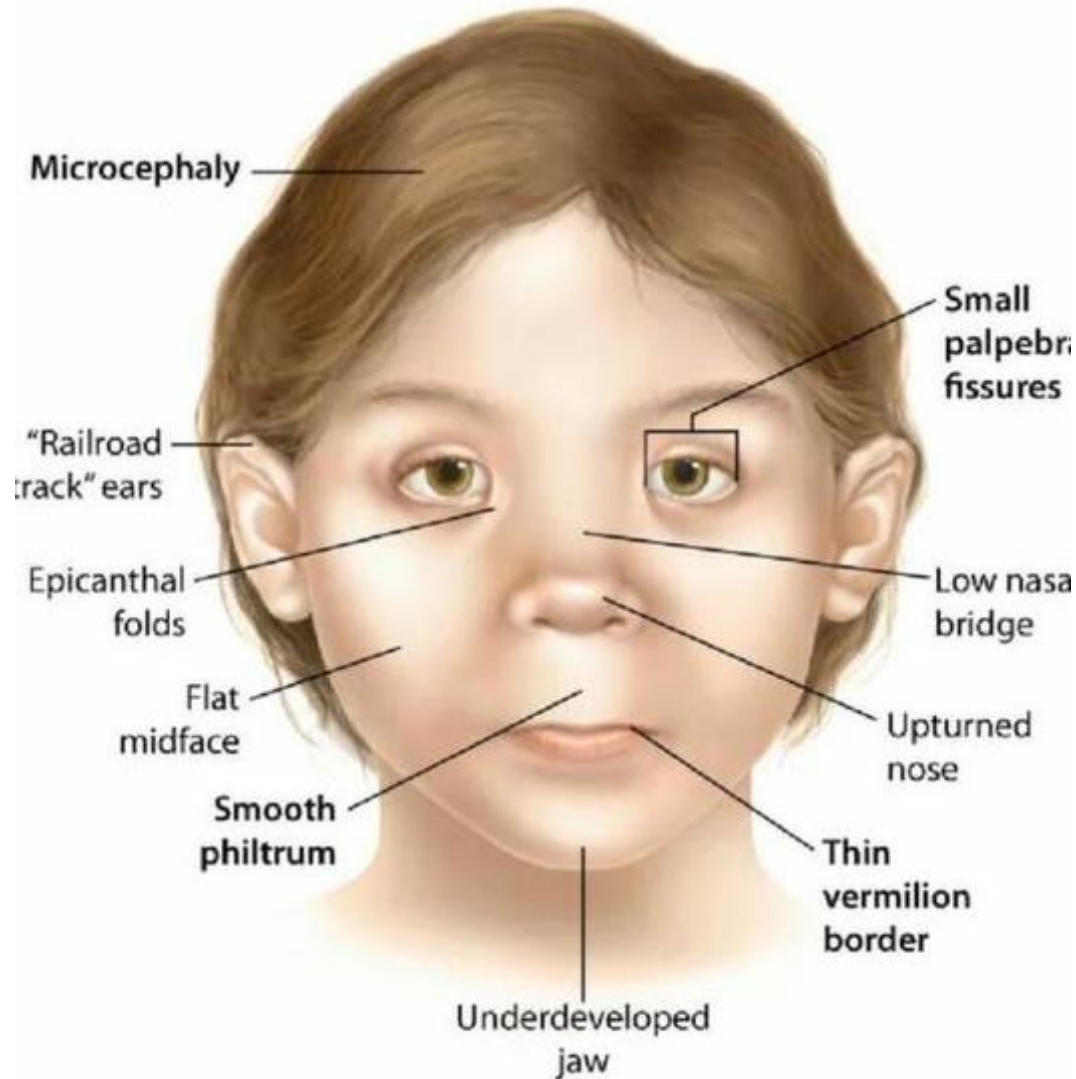
Human teratogenic agents

Alcohol

- The **fetal alcohol syndrome**: craniofacial dysmorphism, intrauterine and postnatal growth retardation, retarded psychomotor and impaired intellectual development, and other nonspecific major and minor abnormalities
- Cleft palate and cardiac anomalies may also occur

craniofacial dysmorphism of FAS

Fetal alcohol syndrome



Thalidomide

- morning sickness, nausea (1956 to 1961)
- offspring malformations mainly *amelia* or *phocomelia*
- Other malformations included: absence of the auricles with deafness, defects of the muscles of the eye and face, and malformations of the heart, bowel, uterus, and the gallbladder
- The compound was *withdrawn* from the market in 1961 after about 10,000 cases had occurred
- Currently= Leprosy, arthritis, multiple myeloma
- Needs strict f/up(pregnancy test, use of contraception)

Birth defects

caused by use of thalidomide



10,000 cases

Diethylstilbestrol

- Widely used in 1940s to the 1970s in the US
- Used for threatened abortion
- **Female offsprings-**
 - ✓ **Vaginal adenocarcinoma** (in 1970's female offspring at age 15 – 22yrs)
 - ✓ **Infertility, Miscarriage** (in 22-58% females due to developed uterine, cervical or tubal changes)
- **Male offspring** - Cryptorchidism, small testes etc...

Isotretinoin (Accutane)

- It is a member of a family of drugs called retinoids, which are related to vitamin A
- It is approved to treat serious forms of acne
- Its use in pregnancy may lead to birth defects
 - ✓ facial malformations, heart defects, and mental retardation

Anticonvulsants

- Women with epilepsy 5-10% risk of anomalies
- Range of anomalies varies with drugs
- Risk increases with numbers and dose of drugs
- *Carbamazepine, phenytoin, valproic* – teratogenic

Others -- ACEI/ARB, lithium, Folate antagonist,
Misoprostol, Tobacco smoke...

Human Developmental Toxicants

❑ Infections

- Cytomegalovirus
- Parvovirus B-19
- Rubella virus
- Syphilis
- Toxoplasmosis
- Varicella virus
- Encephalitis virus
- Zika virus

❑ Radiation

- Therapeutic Radiation
- Radioiodine
- Atomic fallout

❑ Maternal trauma and metabolic imbalances

- Alcoholism
- Cretinism
- Diabetes
- Folic acid deficiency

continued

Drugs/Chemicals

Androgenic chemicals

Angiotensin converting enzyme inhibitors

Captopril, enalapril

Antibiotics

Tetracyclines

Anticancer drugs

Aminopterin, methylaminopterin,
cyclophosphamide, busulfan

Anticonvulsants

Diphenylhydantoin, trimethadione,
valproic Acid

Antithyroid drugs

Methimazole

Chelators

Penicillamine

Chlorobiphenyls

Cigarette smoke

continued

Cocaine

Coumarin anticoagulants (warfarin)

Ethanol

Ethylene oxide

Fluconazole, high dosage

Diethylstilbestrol

Iodides

Lithium

Metals

 Mercury (organic), lead

Methylene blue via intraamniotic injection

Misoprostol

Retinoids

 13-*cis*-retinoic acid, Etretinate

Thalidomide

Toluene abuse



THANK YOU!